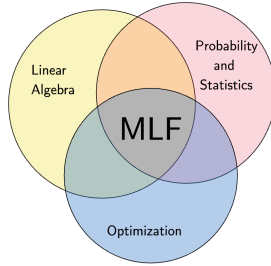


# Week-3, Session-1



Karthik Thiagarajan

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## 1. Vector spaces

A vector space is a set  $V$  along with two operations,  $+$  and  $\cdot$ , addition and scalar multiplication that obeys certain axioms:

Addition,  $+$

- Commutativity:  $x + y = y + x$
- Associativity:  $x + (y + z) = (x + y) + z$
- Additive identity  $0 \in V$ ,  $x + 0 = x$
- Additive inverse:  $x + (-x) = 0$

Scalar multiplication,  $\cdot$

- Multiplicative identity,  $1 \cdot x = x$
- Associativity,  $a \cdot (b \cdot x) = (ab) \cdot x$
- Distributivity:
  - $(a + b) \cdot x = a \cdot x + b \cdot x$
  - $a \cdot (x + y) = a \cdot x + a \cdot y$

Examples of vector spaces:

- $\mathbb{R}^n$ : the set of all  $n$  – tuples
  - $\mathbb{R}^n = \mathbb{R} \times \mathbb{R} \times \dots \times \mathbb{R}$
  - $(x_1, \dots, x_n) \in \mathbb{R}^n$ , where  $x_i \in \mathbb{R}$
  - $\mathbb{R}^2$ , the plane
  - $\mathbb{R}^3$ , the space
- $\mathbb{C}^n$ : the set of all  $n$  – tuples
  - $\mathbb{C}^n = \mathbb{C} \times \mathbb{C} \times \dots \times \mathbb{C}$
  - $\mathbb{C}$  is the set of all complex numbers
  - $(x_1, \dots, x_n)$ , where  $x_i \in \mathbb{C}$

–  $\mathbb{C}^2$

•  $\cdot (2 + 3i, 5i)$

•  $i^2 = -1$  or  $i = \sqrt{-1}$

## 2. Subspaces

Let us consider  $\mathbb{R}^2$ . Is  $\{(0, 1), (1, 0), (0, 0)\}$  a subspace of  $\mathbb{R}^2$ ?

No. It is not.

- First, a subspace should be a subset of a vector space.
- Secondly, a subspace should be a vector space in its own right.

How do you verify if a subset  $S$  is a subspace of a  $\mathbb{R}^2$ ?

- $(0, 0) \in S$
- $S$  should be closed under addition and scalar multiplication

What does closure mean?

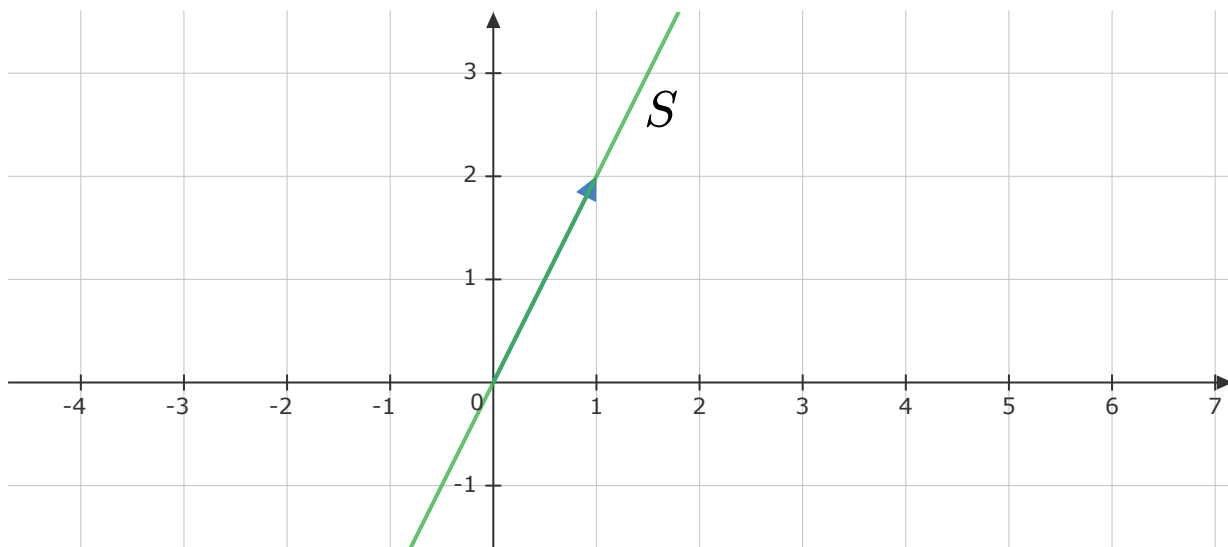
- If you add two vectors, the result should also be in the set.
- If you multiply a vector by a scalar, the result should be in the set.

$$S = \{(0, 1), (1, 0), (0, 0)\}$$

$S$  is not a subspace because  $(0, 1) + (1, 0) = (1, 1) \notin S$

Let's look at some subspaces of  $\mathbb{R}^2$

- all lines passing through the origin are subspaces of  $\mathbb{R}^2$
- $S = \{\alpha \cdot (1, 2) : \alpha \in \mathbb{R}\}$  is a subspace of  $\mathbb{R}^2$



Verifying if  $S$  is closed under addition:

$$\alpha \cdot (1, 2) + \beta \cdot (1, 2) = (\alpha + \beta) \cdot (1, 2) = \gamma \cdot (1, 2)$$

Every vector space has two trivial subspaces. For example,  $\mathbb{R}^2$ :

- $\{(0, 0)\}$
- $\mathbb{R}^2$

### 3. Linear independence and span

#### 3.1. Span

Span of  $S$  is the set of all possible linear combinations of vectors in  $S$ . For example:

$$S = \{(1, 1), (2, 3)\}$$

The following is one linear combination:

$$2 \cdot (1, 1) - 5(2, 3)$$

The span is the set of all possible linear combinations:

$$\text{span}(S) = \{\alpha \cdot (1, 1) + \beta \cdot (2, 3) : \alpha \in \mathbb{R}, \beta \in \mathbb{R}\}$$

In general, if

$$S = \{v_1, \dots, v_n\}$$

$$\text{span}(S) = \{\alpha_1 \cdot v_1 + \dots + \alpha_n v_n : \alpha_i \in \mathbb{R}\}$$

$\mathbb{R}$  or  $\mathbb{C}$  are fields

Properties of span:

- $\text{span}(S)$  is a subspace
- $\text{span}(S)$  is the smallest subspace containing  $S$
- $\text{span}(\{ \}) = \{0\}$ , the trivial subspace

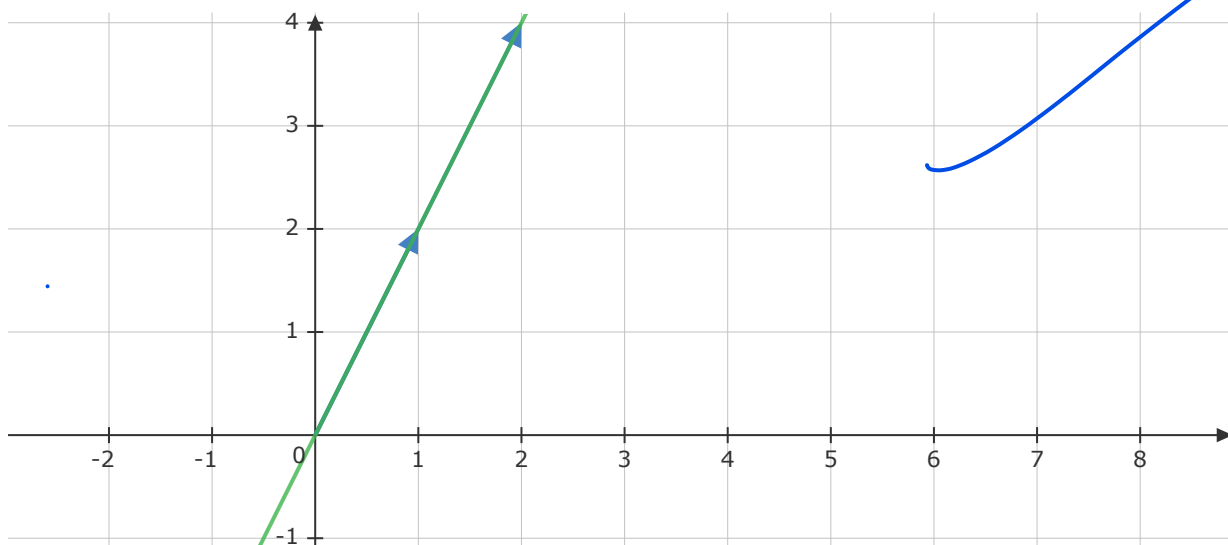
## 3.2. Linear Independence

Let's look at the following vectors:

$$u = (1, 1)$$

$$v = (2, 2)$$

We can write  $v = 2u$ . If you have two vectors  $u, v \in \mathbb{R}^2$ , they are called linearly dependent if one vector can be written as a multiple of the other.  $v = \alpha \cdot u$  or  $u = \alpha \cdot v$  for some  $\alpha \in \mathbb{R}$ , then  $\{u, v\}$  is linearly dependent.



Now consider these vectors in  $\mathbb{R}^3$ :

$$u = (1, 0)$$

$$v = (0, 1)$$

$$w = (1, 1)$$

Notice  $w = u + v$ . We say that  $\{u, v, w\}$  is linearly dependent.

We say that a set of vectors  $S = \{v_1, \dots, v_n\}$  is linearly dependent if there is some vector  $v_i \in S$  that is in the span of the remaining vectors.

$$v_i = \alpha_1 v_1 + \dots + \alpha_{i-1} v_{i-1} + \alpha_{i+1} v_{i+1} + \dots + \alpha_n v_n$$

There is alternative way for defining dependence/independence. First you take any linear combination and set it to zero:

$$\alpha_1 v_1 + \dots + \alpha_n v_n = 0$$

Is there some choice of  $\alpha_i$  that will give me zero? This will always happen when  $\alpha_1 = \dots = \alpha_n = 0$ . Linear independence demands that there is no linear combination other than the trivial one which will result in the zero vector.

$$u = (1, 0)$$

$$v = (0, 1)$$

$$w = (1, 1)$$

$$\alpha_1(1, 0) + \alpha_2(0, 1) + \alpha_3(1, 1) = (0, 0)$$

- $\alpha_1 = \alpha_2 = \alpha_3 = 0$
- $\alpha_1 = 1, \alpha_2 = 1, \alpha_3 = -1 \rightarrow$  non-trivial combination

The moment you have a non-trivial combination, the set of vectors becomes dependent.

A set  $S = \{v_1, \dots, v_n\}$  is **linearly independent** if

$$\alpha_1 v_1 + \dots + \alpha_n v_n = 0 \implies \alpha_i = 0, \quad 1 \leq i \leq n$$

If  $0 \in S$  then,  $S$  is linearly dependent.

### 3.3. Spanning set

A spanning set  $S$  of a vector space  $V$  is a set whose span is the entire vector space, meaning,  $\text{span}(S) = V$ .  $S$  is said to span  $V$ . For example,  $\{(1, 0), (0, 1)\}$  spans  $\mathbb{R}^2$ ,  $\{(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 2, 3)\}$  spans  $\mathbb{R}^3$ .

## 4. Basis and dimension

A basis of a vector space  $V$  is a linearly independent spanning set.

Therefore, if  $B$  is a basis for  $V$ :

- $B$  has to be linearly independent
- $\text{span}(S) = V$

Examples:

- $B = \{(1, 0), (0, 1)\}$  is a basis for  $\mathbb{R}^2$
- $B = \{(1, 0, \dots, 0), \dots, (0, 0, \dots, 1)\}$  is a basis for  $\mathbb{R}^n$
- The above bases are examples of standard bases

Properties/Observations

- **Basis is not unique**
  - $\{(1, 2), (2, 1)\}$  is another basis for  $\mathbb{R}^2$
- **Minimally spanning set:** If you remove a vector from a basis, it is no longer spanning set
- **Maximally linearly independent set:** If you add a vector to a basis, it is no longer linearly independent
- Given any linearly independent set, you can extend it to form a basis:
  - $B' = \{(1, 0, 0), (0, 0, 1)\}$ ,  $B = B' \cup \{(0, 1, 0)\}$
- Given any spanning set, you can shrink it to form a basis:

- $B' = \{(1, 0), (0, 1), (2, 3), (5, 6)\}$
- Throw any two vectors out to get a basis
- 0 can never be an element of a basis
- All bases for a vector space are of the same size

The dimension of a vector space is the number of vectors in the basis.

That is,  $\dim(V) = |B|$ . In this course we will study what are called finite dimensional vector spaces.

Does anyone know an infinite dimensional vector space? The set of all polynomials is an example.

$$a_0 + a_1x + \dots + a_nx^n \in \mathcal{P}$$

$$B = \{1, x, x^2, \dots\}$$

Convention, the basis for the trivial subspace  $\{0\}$  is the empty set  $\{\}$ . Therefore,  $\dim\{0\} = 0$ .  $\{\}$  is a linearly independent spanning set for  $\{0\}$  and therefore a basis.

An example of a one-dimensional vector space would be any line passing through the origin in  $\mathbb{R}^2$ .

[https://en.wikipedia.org/wiki/Vacuous\\_truth](https://en.wikipedia.org/wiki/Vacuous_truth)

## 5. Dot product

So far what all things can we do with vectors?

- add
- scale them by scalars

Why do we need dot products? To do geometry.

- compute lengths of vectors
- compute angles between vectors

$$\langle x, y \rangle = x^T y$$

$$x = (x_1, x_2), y = (y_1, y_2)$$

$$x^T y = x_1 y_1 + x_2 y_2$$

Properties:

- non-negativity:
  - $x^T x \geq 0$
  - $x^T x = 0$  if and only if  $x = 0$
  - $x_1^2 + x_2^2 \geq 0$
- linearity
  - $(x + y)^T z = x^T z + y^T z$
  - $(\alpha \cdot x)^T y = \alpha \cdot (x^T y)$
- symmetry
  - $x^T y = y^T x$

**Note:** It is actually "conjugate symmetry" for complex vector spaces. In complex vector spaces, simple symmetry doesn't work for the complex dot product.

## 6. Norm and angle

The norm of a vector is the length of the vector. We have what is called the dot-product induced norm or the norm induced by the dot product.

$$||x|| = \sqrt{x^T x}$$

The angle between two vectors  $x, y$  in  $\mathbb{R}^2$  is given by:

$$\theta = \cos^{-1} \left( \frac{x^T y}{||x|| \cdot ||y||} \right)$$

$$\cos \theta = \frac{x^T y}{||x|| \cdot ||y||}$$

The above relation is often called the cosine similarity in the ML literature. It measures the similarity between two vectors. Vectors are features or attributes representing data. For example, let's say you are describing a house:

$$x = \begin{bmatrix} \text{area} \\ \text{number of rooms} \end{bmatrix} = \begin{bmatrix} 1000 \\ 3 \end{bmatrix}$$

$$y = \begin{bmatrix} \text{area} \\ \text{number of rooms} \end{bmatrix} = \begin{bmatrix} 1200 \\ 2 \end{bmatrix}$$

Cosine similarity measures the similarity between two houses:

$$\frac{x^T y}{||x|| \cdot ||y||}$$

What is the range of values of the above expression?  $[-1, 1]$ . In general, the higher the cosine-similarity, more similar the vectors or the data-points (houses) you are comparing. If  $x = y$ , the cosine-similarity will be 1.

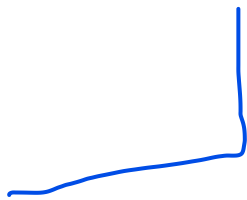
Cauchy-Schwartz inequality:

$$x^T y \leq ||x|| \cdot ||y||$$

How do you convert  $(1, 2, 3)$  into a unit-norm vector?

- First find the norm.
- Divide the vector by the norm.

$$||(1, 2, 3)|| = \sqrt{1^2 + 2^2 + 3^2}$$

$$\frac{(1, 2, 3)}{\sqrt{14}}$$


## 7. Orthogonality

Two vectors  $u, v$  are orthogonal (perpendicular) if  $u^T v = 0$ . For example, the following pairs are orthogonal:

- $(1, 0) \perp (0, 1) \rightarrow \text{perp}$
- $(1, 2, 1) \perp (-1, 1, -1)$
- $(0, 0, 0) \perp u$  for all  $u \in \mathbb{R}^3$ . In general,  $0 \perp V$

Cosine similarity between two orthogonal vectors is 0. 0 is the only vector that is orthogonal to itself.

## 8. Orthogonality and linear independence

A set of vectors  $S = \{v_1, \dots, v_n\}$  is said to be pair-wise orthogonal if  $v_i^T v_j = 0$  for  $i \neq j$ . If  $S$  is a collection of non-zero orthogonal vectors, then  $S$  is linearly independent.

Take any linear combination and set it to zero. To show:

$$\alpha_1 v_1 + \dots + \alpha_n v_n = 0 \implies \alpha_i = 0$$

$$v_i^T (\alpha_1 v_1 + \dots + \alpha_n v_n) = v_i^T 0$$

$$\alpha_1 v_i^T v_1 + \dots + \alpha_{i-1} v_i^T v_{i-1} + \alpha_i v_i^T v_i + \dots + \alpha_n v_i^T v_n = 0$$

$$\alpha_i ||v_i||^2 = 0 \implies \alpha_i = 0$$

Therefore, we have  $\alpha_1 = \dots = \alpha_n = 0$  and that shows that  $S$  is linearly independent.

Orthogonal sets are linearly independent provided the set doesn't have the zero vector.

## 9. Orthonormal basis

$S = \{v_1, \dots, v_n\}$  is an orthonormal basis for  $\mathbb{R}^n$  if:

$$v_i^T v_j = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

The vectors in an orthonormal basis are pair-wise orthogonal and have unit norm. For example:

- $\{(1, 0), (0, 1)\}$  is an orthonormal basis for  $\mathbb{R}^2$
- $\left\{ \frac{(3, 4)}{5}, \frac{(-4, 3)}{5} \right\}$  is an orthonormal basis for  $\mathbb{R}^2$

Note that  $\{(3, 4), (-4, 3)\}$  is an orthogonal basis but not orthonormal.

Now, any vector in  $\mathbb{R}^n$  can be expressed as:

$$v = (v^T v_1)v_1 + \dots + (v^T v_n)v_n$$

For example, consider  $B = \left\{ \frac{(3, 4)}{5}, \frac{(-4, 3)}{5} \right\}$ . How do you represent  $(1, 2)$  in this basis?

$$v = (1, 2), \quad \begin{aligned} v_1 &= \frac{(3, 4)}{5} \\ v_2 &= \frac{(-4, 3)}{5} \end{aligned}$$

$$v = \alpha_1 v_1 + \alpha_2 v_2$$

$$\alpha_1 = v^T v_1$$

$$\alpha_2 = v^T v_2$$

$$= \frac{1}{5} \cdot [1 \ 2] \begin{bmatrix} 3 \\ 4 \end{bmatrix},$$

$$= \frac{1}{5} \cdot [1 \ 2] \begin{bmatrix} -4 \\ 3 \end{bmatrix}$$

$$= \frac{11}{5}$$

$$= \frac{2}{5}$$

$$(1, 2) = \frac{11}{5} \cdot \left[ \frac{(3, 4)}{5} \right] + \frac{2}{5} \cdot \left[ \frac{(-4, 3)}{5} \right]$$

Verify:

$$\frac{(33 - 8, 44 + 6)}{25} = \frac{(25, 50)}{25} = (1, 2)$$

## 10. Four Fundamental Subspaces

A matrix  $A$  of shape  $m \times n$  has four subspaces associated with it:

- Column space
- Row space
- Nullspace
- Left nullspace

### 10.1. Column space

$$A = \begin{bmatrix} | & & | \\ a_1 & \cdots & a_n \\ | & & | \end{bmatrix}$$



The column space is the span of the columns:

$$\mathcal{C}(A) = \text{span}\{a_1, \dots, a_n\}$$

- The column space is a subspace of  $\mathbb{R}^m$
- The dimension of the column space is called column rank

## 10.2. Row space

$$A = \begin{bmatrix} - & a_1^T & - \\ & \vdots & \\ - & a_m^T & - \end{bmatrix}$$

$$\mathcal{R}(A) = \text{span}\{a_1^T, \dots, a_m^T\}$$

- The row space is a subspace of  $\mathbb{R}^n$
- The dimension of the row space is called the row rank
- The row rank is equal to the column rank. We will just use the term "rank" of the matrix.
- We have  $\mathcal{C}(A^T) = \mathcal{R}(A)$  because the operation of transpose swaps the rows and columns. What was a row before now becomes a column and vice-versa.

### Rank of a matrix

- The number of linearly independent columns.
- The number of linearly independent rows.

## 10.3. Null space

The null space of a matrix  $A$  is the set of all solutions to the homogeneous system  $Ax = 0$ .

$$\mathcal{N}(A) = \{x : Ax = 0, x \in \mathbb{R}^n\}$$

The dimension of the null space is called nullity. It is known that:

$$\text{rank} + \text{nullity} = \text{number of columns}$$

This is the rank-nullity theorem.

## 10.4. Left null space

The set of all solutions to the equation  $y^T A = 0$  is the left null space:

$$\mathcal{N}(A^T) = \{y : A^T y = 0, y \in \mathbb{R}^m\}$$

$$y^T A = 0 \implies (y^T A)^T = 0 \implies A^T y = 0$$

Remember the following pairing:

- Null space and the row space are subspaces of  $\mathbb{R}^n$
- Column space and the left null space are subspaces of  $\mathbb{R}^m$

## 10.5. Algorithm to compute the four subspaces

We will do this with the help of an example:

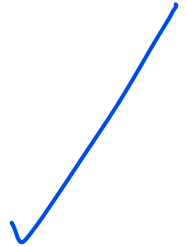
$$A = \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 1 & 1 & 4 & 3 \end{bmatrix}$$

Row space

- Reduce the matrix to its row echelon form using elementary row operations:

- Scale a row by a **non-zero** constant
- Swap rows
- Add a constant times one row to **another** row

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 1 & 1 & 4 & 3 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_1} \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 0 & 1 & 3 & 1 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 0 & 1 & 3 & 1 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2} \begin{bmatrix} \mathbf{1} & 0 & 1 & 2 \\ 0 & \mathbf{1} & 3 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

- The first non-zero element in each row is the pivot.
- The non-zero rows of the row-echelon matrix is a basis for the row space.
- Every row operation doesn't disturb the row space. The row space is preserved across all row operations.

Row space is given by  $\text{span}\{(1, 0, 1, 2), (0, 1, 3, 1)\}$

We are trying to show the non-zero rows in the row-echelon form are linearly independent and span the row space.

$$\begin{bmatrix} \mathbf{1} & 0 & 1 & 2 \\ 0 & \mathbf{1} & 3 & 1 \end{bmatrix} \rightarrow$$

$$\alpha(1, 0, 1, 2) + \beta(0, 1, 3, 1) = (0, 0, 0, 0)$$

$$(\alpha, \beta, \alpha + 3\beta, 2\alpha + \beta) = (0, 0, 0, 0)$$

$$\implies \alpha = \beta = 0$$

Why is this case? The pivot in every row is to the right of the pivot in the

previous row.

## Column space

$$A = \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 1 & 1 & 4 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 1 & 1 & 4 & 3 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_1} \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 0 & 1 & 3 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 0 & 1 & 3 & 1 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2} \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

- Identify the pivot columns: the columns that contain the pivots
- Go back to the original matrix and extract the columns corresponding to the pivot column
- These columns form a basis for the column space

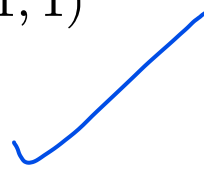
$$\mathcal{C}(A) = \text{span}\{(1, 0, 1), (0, 1, 1)\}$$

Verifying that we have a spanning set for the column space.

$$(1, 3, 4) = 1(1, 0, 1) + 3(0, 1, 1)$$

$$(2, 1, 3) = 2(1, 0, 1) + 1(0, 1, 1)$$

So far we have found out the rank is 2.



## Null space

$$A = \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 1 & 1 & 4 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 1 & 1 & 4 & 3 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_1} \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 0 & 1 & 3 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 0 & 1 & 3 & 1 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2} \begin{bmatrix} \mathbf{1} & 0 & 1 & 2 \\ 0 & \mathbf{1} & 3 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Homogeneous system:

We want to find a basis for the nullspace.

$$\begin{bmatrix} \mathbf{1} & 0 & 1 & 2 \\ 0 & \mathbf{1} & 3 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \mathbf{x}_3 \\ \mathbf{x}_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

- $x_1, x_2$  are the pivot variables
- $x_3, x_4$  are the free variables

Set  $x_3 = 1, x_4 = 0$

$$x_2 = -3, x_1 = -1$$

Therefore, one vector in the basis is  $(-1, -3, 1, 0)$ .

Set  $x_3 = 0, x_4 = 1$

$$x_2 = -1, x_1 = -2$$

Therefore, the other vector in the basis is  $(-2, -1, 0, 1)$ .

Finally, a basis for the nullspace of  $A$  is:

$$\{(-1, -3, 1, 0), (-2, -1, 0, 1)\}$$

and

$$\mathcal{N}(A) = \text{span}\{(-1, -3, 1, 0), (-2, -1, 0, 1)\}$$

Verify if these two are actually in the nullspace.

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 1 & 1 & 4 & 3 \end{bmatrix} \begin{bmatrix} -1 \\ -3 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \\ 1 & 1 & 4 & 3 \end{bmatrix} \begin{bmatrix} -2 \\ -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

- The row space is orthogonal to the null space.
- The column space is orthogonal to the left null space.

